A Project Report on

MULTI-AGENT PPO-BASED RELAY SELECTION AND POWER ALLOCATION FOR COOPERATIVE NOMA NETWORK

submitted in partial fulfilment of the requirements for the course

MACHINE LEARNING
25SC2107E

in

A.Y.: 2026-27

by

G. Harshith **roll 2520040079**

V. Shivadarahas **roll 2520040038**

B. Pavan **roll 2520040102**

Under the Esteemed guidance of

Mr. N Lava Prasad

Assistant Professor,

Department of ECE



(DEEMED TO BE UNIVERSITY)

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF **ELECTRONICS & COMMUNICATION ENGINEERING**

Abstract:

This project investigates intelligent relay selection and transmission power allocation in cooperative Non-Orthogonal Multiple Access (NOMA) wireless networks. Wireless networks must support multiple users while maintaining suitable data rates, signal quality, reliability, and efficient use of limited transmission power. Network conditions change with distance, channel quality, noise, interference, and user requirements. When the direct link between a base station and a user is weak, a relay may assist by forwarding information. The proposed approach investigates reinforcement learning, with Proximal Policy Optimization (PPO) as the intended algorithm, in a virtual wireless-network environment containing a base station, relays, and users. The simulation considers channel quality, signal strength, interference, user requirements, and a total transmission-power budget. A policy observes the current state and selects relay actions, while a power-allocation component distributes power according to a documented rule. A reward function represents the desired balance between communication performance and resource use. The exact reward design and multi-agent coordination must be specified and validated during implementation. Performance is evaluated using throughput, data rate, SINR, BER, outage probability, energy efficiency, and power consumption. The intended comparison methods are random relay selection, fixed relay selection, and equal-power allocation. Python, Google Colab, NumPy, PyTorch, Pandas, and Matplotlib provide a practical implementation environment. This is a software simulation and does not claim measurements from a live 5G/6G network. Numerical conclusions should be drawn only from verified program outputs.

Signature of the Supervisor

Course Instructor

INDEX

Abstract ii

1. Introduction 1
    i. 1.1 Background and Motivation 1
    ii. 1.2 Problem Statement and Objectives 2
2. 2. System Model 5
    i. 2.1 Network Components 6
    ii. 2.2 Observations, Actions and Rewards 7
    iii.
3. 3. Proposed Methodology
    i. 3.1 Reinforcement Learning and PPO 6
    ii. 3.2 Helper Selection and Power Allocation 7
    iii.
4. 4. Performance Evaluation
5. 5. Conclusion and References

1. Introduction

You start writing Introduction Here

Wireless communication is essential for mobile networks, Internet of Things applications, and emerging 5G/6G services. These networks must support multiple users with reliable communication while using limited spectrum and transmission power. Signal quality changes because of distance, obstacles, movement, fading, noise, and interference. When the direct connection between a base station and a user is weak, a relay device can help by receiving and forwarding information. Non-Orthogonal Multiple Access (NOMA) allows multiple users to share a time or frequency resource using non-orthogonal signals. In cooperative NOMA, relay assistance can improve connectivity for users with weak direct links. However, choosing a suitable relay and allocating transmission power are challenging because the best decision depends on current channel conditions and user requirements. This project studies a reinforcement-learning approach using PPO to make relay-selection decisions and support power allocation in a simulated network. The system observes network conditions, evaluates actions through a reward function, and learns over repeated simulation episodes. Its performance is intended to be evaluated using throughput, data rate, SINR, BER, outage probability, energy efficiency, and power consumption.

1.1 BACKGROUND AND MOTIVATION

Wireless communication supports mobile applications, online learning, video calls, connected sensors, and smart devices. As the number of connected devices grows, networks must deliver information reliably while managing limited spectrum and transmission power.

This project focuses on two resource-management decisions: selecting a suitable helper device and determining an appropriate transmission power level. A helper, also called a relay, is an intermediate node that can assist a wireless connection when the direct signal between a base station and a user is weak.

Wireless conditions change over time. A helper that is suitable for one user or one channel condition may not be the best choice in another situation. Fixed rules may not adapt well when a user moves, the channel changes, or interference increases. Reinforcement learning provides a way to learn from repeated interactions with a simulated environment.

The project is intended as a software simulation. It provides a controlled setting to study the decision process, calculate performance metrics, and compare an AI-based approach with simpler methods. Conclusions must remain within the assumptions of the implemented simulation.

Motivation

- Improve the selection of helper devices under changing conditions.
- Use transmission power more carefully.
- Evaluate communication quality and reliability with measurable metrics.
- Compare the proposed method with simple baselines.

1.2 PROBLEM STATEMENT AND OBJECTIVES

In a wireless network with multiple users and candidate helpers, selecting a suitable helper and deciding an appropriate power level are challenging tasks. Link quality can vary with distance, obstacles, channel fading, noise, and interference. A fixed selection or power rule may not adapt to all situations.

The central problem is how an AI-based decision system can use current network observations to choose a suitable helper and manage transmission power in a simulated 5G/6G-style network. The system must be evaluated with clearly defined metrics and compared fairly with baseline approaches.

Objectives

- Create a virtual network containing a base station, helpers, and users.
- Represent signal strength, distance, noise, interference, user requirements, and power availability.
- Investigate PPO-based reinforcement learning for helper selection.
- Apply a documented power-allocation method that respects the total power budget.
- Measure throughput, data rate, SINR, BER, outage probability, energy efficiency, and power consumption.
- Compare with random helper selection, fixed helper selection, and equal-power allocation.
- Generate graphs and tables from actual simulation runs.

1.3 SCOPE AND CONTRIBUTION

The scope is a simulation-based study rather than a live radio-network deployment. Network state, helper conditions, noise, interference, and user requirements are represented numerically in software. This enables repeatable experiments under controlled settings.

The intended contribution is an integrated workflow: generate a network state, select a helper, apply power allocation, calculate communication metrics, obtain feedback, and evaluate the policy. Baseline comparisons provide reference points for assessing whether the learned method is useful.

The model is simplified. If the implementation does not include all physical-layer NOMA mechanisms, such as power-domain signal superposition and successive interference cancellation, it should be described as a simplified NOMA-inspired simulation rather than a complete physical-layer implementation.

Included

- Software simulation of a base station, helpers, and users.
- PPO-based learning experiment, subject to correct implementation.
- Helper-selection and power-allocation decisions.
- Metric calculations, baseline comparisons, and graph generation.

Outside the current scope

- Live cellular-network testing.
- Guarantees of real-world 5G/6G improvement.
- Claims of a fully multi-agent system unless multiple agents and coordination are implemented.

2. SYSTEM OVERVIEW

The simulated network contains a base station, a set of helper devices, and multiple users. The base station is the central transmission point. A helper or relay can assist communication when a direct link is weak. Users represent devices that receive information from the network.

At each simulation step, the model generates or updates conditions such as link quality, signal strength, interference, noise, user requirements, and available transmission power. These values form the observation supplied to the AI policy.

The decision system selects a helper and determines or supports a power-allocation decision. The simulation calculates the communication outcome and metrics. During training, a reward function summarizes the outcome and supplies feedback to the learning algorithm.

The system can be divided into a simulation layer, decision layer, learning layer, and evaluation layer. Separating these components makes the implementation easier to test and allows baseline methods to use the same environment.

Main components

- Network simulator and topology.
- Observation builder.
- Helper-selection policy.
- Power controller.
- Metric calculator.
- Training and evaluation module.

2.1 WIRELESS NETWORK CONCEPTS

A base station is the central transmission point in the simulated network. A helper or relay is an intermediate device that receives and forwards information to assist a user. A relay is not the same thing as transmission power: the relay is a node, while power allocation is a separate decision about how much transmission power is used.

NOMA stands for Non-Orthogonal Multiple Access. It allows multiple users to share a time or frequency resource using non-orthogonal signals. A full power-domain NOMA model may include signal superposition and receiver processing such as successive interference cancellation. The report must match the mechanisms actually implemented in code.

Signal strength describes the level of the useful signal. Noise is unwanted random signal energy, while interference comes from other transmissions or signals. SINR compares useful signal power with interference and noise. A higher SINR generally indicates better signal conditions, although data rate also depends on the communication model.

Simplified communication path

Base station → selected helper/relay → user. This is a simplified representation; the simulation may also include a direct base-station-to-user link.

2.2 NETWORK OBSERVATIONS AND DECISIONS

The AI can make decisions only from the information supplied in its observation vector. Candidate variables include signal strength, distance, channel quality, noise, interference, user data-rate requirements, helper conditions, and available power. The final code should document exactly which variables are used.

A decision may include a helper index identifying the selected relay. If power allocation is learned, the action space must represent a valid power decision. Alternatively, PPO may be used for helper selection while a separate rule handles power allocation. These designs are different and must be described accurately.

The total power constraint must be respected. If power is allocated across N users, the sum of assigned powers should not exceed the configured total budget. The implementation should specify whether power is applied at the base station, at the relay, or across a defined set of transmitters.

Inputs

- Channel or link quality.
- Signal strength and distance.
- Noise and interference.
- User requirements.
- Available power and helper conditions.

Outputs

- Selected helper or relay.
- Power decision or power-rule output.
- Performance metrics.
- Reward value during training.

3. PROPOSED METHODOLOGY

The methodology combines a virtual wireless environment with reinforcement learning. The simulator generates the network configuration and current conditions. The policy receives the observation and selects an action. The environment applies the action, calculates the communication outcome, and returns metrics and a reward.

Training and final evaluation should be separated. Training episodes update the policy. Evaluation episodes measure performance without updating the model. Baselines should be tested under comparable conditions so differences are not caused by inconsistent settings.

Random helper selection chooses a candidate at random. Fixed helper selection always chooses a predetermined helper. Equal-power allocation distributes the configured power budget evenly. The exact combination of helper and power baselines should be documented.

Workflow

- Initialize the network and simulation settings.
- Generate channel and interference conditions.
- Build the observation vector.
- Select a helper and apply power allocation.
- Calculate communication performance.
- Compute reward and update PPO during training.
- Evaluate against baselines and save results.

3.1 REINFORCEMENT LEARNING

Reinforcement learning is a machine-learning approach in which an agent learns by interacting with an environment. The agent observes a state, chooses an action, and receives a reward. Through repeated interactions, it adjusts its policy to increase the expected total reward.

In this project, the environment represents the wireless network. The state contains numerical information about simulated conditions. The action represents a helper-selection decision and, if supported by the implementation, a power-allocation decision. The reward measures how well the action meets the project objectives.

The reward function influences what the agent learns. If the reward focuses only on throughput, the policy may use too much power. If power use is penalized too strongly, it may save power while delivering poor service. Reward terms should be scaled and validated rather than chosen without explanation.

Key terms

- Agent: the decision-making learning system.
- Environment: the simulated wireless network.
- State: information available to the agent.
- Action: the selected helper or power decision.
- Reward: a numerical score for the outcome.
- Policy: the rule learned for selecting actions.

3.2 PPO ALGORITHM

PPO stands for Proximal Policy Optimization. It is a policy-gradient reinforcement-learning algorithm that updates a policy using experience while limiting excessively large policy changes. It is often implemented with an actor and a critic.

The actor represents the policy and produces probabilities or parameters for possible actions. The critic estimates the value of a state. Training uses rewards and estimates of advantage to determine whether an action was better or worse than expected. PPO's clipped objective discourages overly large policy changes in one update.

A correct implementation must calculate the PPO objective and update the model parameters. Creating a neural network or generating actions from it is not enough to establish that PPO training is implemented. Training curves and separate evaluation results are needed to assess performance.

Implementation checks

- Define observation and action spaces.
- Collect observations, actions, rewards, and log probabilities.
- Calculate returns and advantages.
- Apply the PPO objective and value loss.
- Perform optimizer updates.
- Track training reward and evaluate the policy separately.

3.3 MULTI-AGENT DESIGN

The presentation describes a multi-agent approach. A genuine multi-agent implementation must define each agent, its observations, its actions, and how it coordinates with other agents. Having multiple users does not automatically make a system multi-agent.

Possible designs include separate agents associated with users or a shared policy that makes decisions for multiple users. These designs have different coordination requirements. The actual code must determine which design is used.

Coordination may be needed to avoid conflicting decisions or exceeding the shared power budget. If the prototype uses one PPO policy for multiple users, it should be described as a shared-policy prototype until a multi-agent architecture is implemented and verified.

Items to document

- Number and role of agents.
- Observation and action space for each agent.
- Whether policy parameters are shared.
- Coordination method and shared constraints.
- How the agent arrangement is evaluated.

3.4 POWER ALLOCATION AND REWARD DESIGN

Power allocation distributes available transmission power among users or links. It is separate from helper selection. A simple baseline is equal-power allocation: if total power is P_total and there are N users, each receives P_total/N. A proportional rule may distribute power according to link-quality scores, but that remains a heuristic unless the policy learns the power action.

The reward function converts project goals into a numerical training signal. It may reward useful throughput and signal quality while penalizing error, outage, or excessive power use. Terms should be scaled so one metric does not dominate simply because its numerical values are larger.

The final reward formula and weights are implementation choices. They should be documented and tested. The report should also state where power is applied and how the total power constraint is enforced.

Power constraint

The sum of allocated powers must not exceed the configured total power budget.

Reward principle

Encourage useful communication while accounting for reliability and resource use; validate the balance using evaluation results.

4. SYSTEM ARCHITECTURE

The architecture can be divided into modules so each component can be tested independently. The network module stores users, helpers, and the base station. The channel module generates link conditions. The environment handles observations, actions, transitions, and rewards. The PPO module learns the policy, while the evaluation module calculates metrics and compares baselines.

Separating modules avoids mixing simulation assumptions with learning code. The channel model can be changed without rewriting graph generation, and new baselines can be added while retaining the same metric calculations.

The code should use consistent units and document variable meanings. Signal power, noise, and interference must use compatible units in SINR calculations. If values are normalized, they should be labelled as normalized simulation units rather than physical measurements.

Suggested modules

- Network topology and configuration.
- Channel and interference generation.
- Observation and action handling.
- Helper selection and power allocation.
- PPO training and model saving.
- Baselines, evaluation, and result export.

4.1 SOFTWARE REQUIREMENTS

Python is suitable for simulation and machine learning because it provides numerical, neural-network, data-processing, and visualization libraries. Google Colab provides a convenient notebook environment for running experiments without a complex local installation.

NumPy supports numerical calculations, PyTorch can implement neural networks and PPO updates, Pandas organizes metrics into tables and CSV files, and Matplotlib creates training and performance graphs. Gymnasium or a custom environment can define reset and step operations, states, actions, and rewards.

These are recommended tools. The report should not state that a tool or feature is implemented unless it is used by the final code.

Recommended stack

- Python 3.x and Google Colab.
- NumPy for simulation calculations.
- PyTorch for PPO.
- Pandas for result tables and CSV export.
- Matplotlib for graphs.
- Optional: Gymnasium and SciPy.

Hardware requirement

A physical base station, relay, or mobile device is not required for the software-simulation version.

4.2 IMPLEMENTATION WORKFLOW

The first step is to define the simulation configuration, including the number of users and helpers, total power, noise assumptions, and episode length. The next step is to generate the network state and link conditions. These parameters should be kept in a configuration section so they can be changed consistently.

The environment accepts an action, calculates the communication outcome, updates the state, and returns metrics and reward. The training code collects experience and uses PPO to update the policy. After training, the evaluation module runs without updating the policy.

Baselines are evaluated under comparable settings. The program records metrics across multiple runs and calculates summary statistics such as mean and standard deviation. Graphs and CSV files should be generated from the same stored results.

Implementation order

- Test the simulator first.
- Validate metric formulas and power constraints.
- Implement and verify PPO updates.
- Add baseline methods.
- Run separate evaluation episodes.
- Generate final tables and graphs.

5. PERFORMANCE METRICS

Performance metrics judge the behavior of the simulated network. No single metric provides a complete picture. For example, throughput may increase when more power is used, but energy efficiency may not improve. Evaluation should consider speed, signal quality, reliability, and power usage together.

Throughput

The total amount of data successfully delivered across the network over a given interval. Higher is generally desirable.

User data rate

How quickly an individual user receives information. Per-user results can reveal whether an average hides poor service for some users.

SINR

Useful signal power divided by interference plus noise. Higher values generally indicate better signal conditions.

BER

The fraction or probability of transmitted bits received incorrectly. Lower values are better.

Outage probability

The probability that a link fails to meet a required rate or quality threshold. Lower is better.

Energy efficiency

Useful data delivered per unit of energy or power, according to the chosen formula. The definition and units should be stated.

Power consumption

Transmission power used by the simulation. It should be interpreted alongside throughput and reliability.

5.1 BASELINE METHODS

Baseline methods provide reference points for evaluating the learning-based approach. They should be clearly defined and tested under the same simulation conditions. If methods use different channel samples, power limits, or metric formulas, the comparison may be unfair.

Random helper selection chooses a candidate relay randomly. Fixed helper selection always uses a predetermined helper, even when conditions change. Equal-power allocation divides the total available power evenly among the relevant users or links.

The comparison should identify which part of the system changes. If random helper selection still uses equal power, state that explicitly. Otherwise, a performance difference may be incorrectly attributed to the relay-selection method alone.

Baseline definitions

- Random helper: choose an available helper randomly.
- Fixed helper: choose a predetermined helper.
- Equal power: divide the configured power budget equally.
- PPO-based method: use the trained policy for actions actually learned by the implementation.

5.2 EXPERIMENTAL PROCEDURE

Record the number of users and helpers, total power budget, noise and interference assumptions, training episodes, learning rate, reward formula, random seed, and evaluation episode count. These settings support reproducibility.

Train the PPO policy using training episodes. Then evaluate the trained policy on separate episodes where model parameters are not updated. Run baselines under comparable settings and, where possible, use the same network scenarios to reduce random variation.

Repeat evaluation and calculate averages and variability for each metric. Generate graphs directly from the stored results. If a metric does not improve, report it honestly and discuss possible reasons rather than inserting expected improvements.

Quality checks

- Verify the notebook runs without errors.
- Confirm PPO optimizer steps update model parameters.
- Check allocated power never exceeds the budget.
- Check metrics are finite and use consistent units.
- Verify graph values match CSV output.
- Label all outputs as simulation results.

6. RESULTS AND DISCUSSION

This section must be finalized using actual outputs from the implemented simulation. The project proposes comparisons with random helper selection, fixed helper selection, and equal-power allocation. Numerical improvement percentages should not be reported until the code has been executed and the results checked.

The training graph can show how reward changes during training, but it does not by itself prove that wireless performance improved. The trained policy must be evaluated on separate episodes. Throughput, SINR, BER, outage probability, energy efficiency, and power consumption should be compared using the same metric definitions.

Results may show trade-offs. A method may improve throughput while increasing power consumption, or save power while reducing data rate. The discussion should explain these trade-offs rather than assuming that every metric improves at once.

Results reporting

Insert the actual values produced by the final Colab evaluation. Do not use invented numbers or percentages.

6.1 GRAPH INTERPRETATION

A training-convergence graph plots episode number against reward or return. A stable trend may suggest learning progress, but a noisy curve is common in reinforcement learning. It should be interpreted alongside independent evaluation metrics.

A throughput graph compares total data delivered by each method. An SINR graph compares signal quality. BER and outage graphs are generally better when their values are lower. Energy-efficiency graphs should use a clearly defined data-per-energy formula.

Power-consumption graphs show whether a method uses more or less transmission power. Lower power is not automatically better if throughput or reliability falls significantly. Graphs should have descriptive titles, labelled axes, units, legends, and repeated-run variability where available.

Suggested figures

- PPO training reward versus episode.
- Throughput comparison.
- SINR comparison.
- Energy-efficiency comparison.
- BER and outage comparison.
- Power-consumption comparison.

7. LIMITATIONS

The project is a software simulation, so conclusions depend on the assumptions used by the environment. It does not reproduce every factor in a real cellular network, such as detailed antenna patterns, scheduler behavior, hardware impairments, mobility standards, or all physical-layer decoding processes.

The channel and interference models may be simplified. If the simulation uses generated signal-quality scores rather than a complete propagation model, results should be described as outputs of that simplified model. Likewise, simplified BER or throughput formulas should not be treated as validated physical-layer measurements.

The reward function affects the behavior learned by PPO. Poor scaling or unsuitable weights can cause the agent to prioritize one metric at the expense of others. Training may also be sensitive to random seeds, learning rate, network size, and number of episodes.

A multi-agent claim requires the implementation to define multiple agents and their coordination. A single policy used for many users is not necessarily a fully multi-agent system.

Reducing limitations

- Use documented channel assumptions.
- Validate formulas with hand-calculated examples.
- Run multiple seeds and report variability.
- Document reward terms and agent roles.
- Avoid claims beyond the simulation evidence.

7.1 FUTURE ENHANCEMENTS

Future work can improve the simulator with more detailed propagation, fading, interference, and user-mobility models. A more complete cooperative-NOMA implementation could include power-domain signal superposition and receiver processing such as successive interference cancellation, where appropriate.

PPO training can be improved by testing different learning rates, clipping parameters, reward scaling, network sizes, and episode counts. Evaluation should use repeated runs and a fixed test configuration.

A clearly defined multi-agent structure is another possible enhancement. Each agent's observation, action, reward, and coordination strategy should be documented. Shared power constraints must be enforced even when several agents make decisions.

A web dashboard can be connected to the Python backend to display the simulated network, selected helpers, power settings, metrics, and baseline charts. It should display actual model outputs and label all data as simulated.

Suggested future plan

- Verify simulator and metric formulas.
- Complete and test PPO training.
- Define and evaluate multi-agent coordination.
- Run larger and repeated experiments.
- Connect a dashboard and finalize documentation.

8. CONCLUSION

This project investigates AI-based helper selection and power allocation for 5G/6G-style wireless networks. A helper or relay can assist communication when a direct link is weak, while power allocation determines how much transmission power is used. These are separate but related resource-management decisions.

The proposed approach uses reinforcement learning, with PPO as the intended algorithm, to learn from a simulated environment. The environment represents a base station, helpers, users, signal conditions, interference, user requirements, and power constraints. Performance is evaluated through throughput, data rate, SINR, BER, outage probability, energy efficiency, and power consumption.

The method should be compared with random helper selection, fixed helper selection, and equal-power allocation under fair and repeatable conditions. A reliable conclusion requires verified code execution, a correct PPO training loop, clear documentation of agent roles and reward design, and actual evaluation results.

The project provides a framework for studying AI-assisted wireless resource management. Future work can include more detailed channel modelling, explicit multi-agent coordination, larger experiments, and a web dashboard. Claims should remain within the limits of the implemented simulation.

REFERENCES

- Project presentation: “AI-Based Smart Helper Selection and Power Allocation for 5G/6G Wireless Networks,” Machine Learning course, 25SC2107E, A.Y. 2026–27.
- J. Schulman et al., “Proximal Policy Optimization Algorithms,” arXiv:1707.06347, 2017.
- R. S. Sutton and A. G. Barto, Reinforcement Learning: An Introduction, 2nd ed., MIT Press, 2018.
- T. M. Cover and J. A. Thomas, Elements of Information Theory, 2nd ed., Wiley, 2006.